



Engineering Degree Project

Title of your degree project
- *Optional subtitle*



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Subject: Computer Science

1 Choice and explanation of dataset

The dataset that I chose for this assignment contains hourly air pollution and weather measurements collected over several years at the US embassy in Beijing China [1]. It is meant to show how meteorological weather can relate to PM2.5 concentrations. Each row contains one hour of observation at a single fixed monitoring location.

This is how the head of the data looks

```
4DT912-advanced-infoviz/A2 on P main via v3.12.3
> head LSTM-Multivariate_pollution.csv
date,pollution,dew,temp,press,wnd_dir,wnd_spd,snow,rain
2010-01-02 00:00:00,129.0,-16,-4.0,1020.0,SE,1.79,0,0
2010-01-02 01:00:00,148.0,-15,-4.0,1020.0,SE,2.68,0,0
2010-01-02 02:00:00,159.0,-11,-5.0,1021.0,SE,3.57,0,0
2010-01-02 03:00:00,181.0,-7,-5.0,1022.0,SE,5.36,1,0
2010-01-02 04:00:00,138.0,-7,-5.0,1022.0,SE,6.25,2,0
2010-01-02 05:00:00,109.0,-7,-6.0,1022.0,SE,7.14,3,0
2010-01-02 06:00:00,105.0,-7,-6.0,1023.0,SE,8.93,4,0
2010-01-02 07:00:00,124.0,-7,-5.0,1024.0,SE,10.72,0,0
2010-01-02 08:00:00,120.0,-8,-6.0,1024.0,SE,12.51,0,0
```

Figure 1.1: Head of the data

As we can see we have a mix of different types of data. We have a datetime which is a sort of ordered continuous time axis, we have numeric and ratioed data as well as nominal wind-direction data.

2 why is this interesting and what can we do with it?

Because the original purpose of this dataset is to help understand and forecast PM2.5 pollution. I think that a use case would be for environmental analysts or urban planners and policy makers that are interested in how pollution evolves over time and which weather situations are associated with unhealthy concentrations.

Some typical questions that I would like to answer with a visual analytics tool could be:

- How do PM2.5 levels vary during different time frames like by year, month or day?
- How do combinations of temperature, wind and rain affect PM2.5?
- Can we discover patterns in the data that correspond to "clean" to "polluted conditions"?

Some domain specific tasks I would like to perform with the data could be:

- inspect long term trends and seasonal cycles in PM2.5 levels. Identify and zoom into specific high-pollution episodes.
- Explore relationships between PM2.5 and variables such as temp, dew point, pressure and wind speed to see which conditions correlate with high or low pollution.
- compare "clean" and "polluted" areas in the data.

Also, because the dataset spans years of hourly measurements, it makes it very dense. Showing everything at once would be overwhelming for a human with a low attention span like me. this is why i will use filters and queries to allow the user to efficiently skim through interesting areas and patterns in the data so that users can focus on specific time periods, conditions and pollution levels. I will also try to use MCV in order to show the user different perspectives of the same data in as few clicks as possible to help the user to get a holistic but valuable insight in to the data.

3 Prototype and conceptual design

For this project i used draw.io to sketch up a wireframe with some concept features that i think would is appropriate for this project and data.

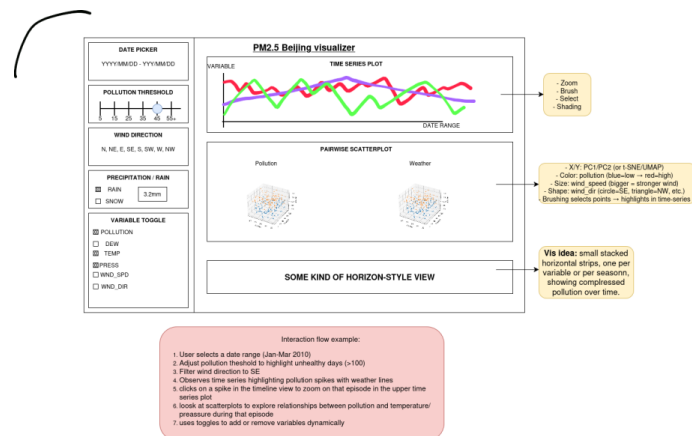


Figure 3.2: Simple wireframe/prototype of the proposed system

As we can see i designed the system in such a way to allow for two main windows. To the left, we have all of the interactive buttons in place to browse, filter and query the data. Since we have dates in the dataset i felt it to be natural to select a range of date in order to allow for the user to look at the data within a specified range. We also have weather variables such as rain and wind so i will also have some kind of slider or simple text-box that will allow for that. Since pollution is also a main variable in the data i want to also allow for it to be changed. At the bottom i will allow for a big list check boxes to allow for the user to view only a subset of variables if they want to see a specific variable, collection of variables or all variables.

Next to the query-menu i have my 3 main graphs/plots that will dynamically change based on the data that has been filtered. The top graph i think could be a time-series plot based on the daterange and variables chosen. The middle plot is a pairwise scatterplot which will show 8D points projected to either 2D or 3D using t-SNE and it will be color, size and shape coded based on pollution, wind-speed, wind-direction variables. I will also make this possible to rotate and zoom. It is pairwise because i want the user to be able to get two different perspectives of the same data without having to enter or exit different plots. This might be ambitious but i would like to try it out and see if i can make something insightful out of having dual scatterplots like this.

The third idea i want to have some kind of horizon-style graph which shows a density view of daily pollution cycles. Im not entirely sure how to implement this but my idea is to group the data by the hour (00-23) and for each hour, show the distribution across

all days and color cade based on seasons or filter by sliders. I think this could be a cool complement to the top time series plot that i already have.

One thing to make clear is i really want the system to be performant and fast so that wen data is queried the plots quickly dynamically transform and change based on the query. That way the entire system will feel responsive and interactive and allow the user to make different comparisons by seeing the same data but from different views and i think will be the most powerful part of te project.

4 Proposed tech stack

I think for this project i will use the same tech stack as i used for the last infovis course which is a simple locally hosted webapp in raw HTML using CDN in order to get a prettier UI and to get the dependencies that i need like Plotly or D3. I plan on having everything on the same page so this is why i feel like i dont need to use compoonents or page-routing from react or vue.

For backend and potentiall processing or pre-processing i will use Python because it has a lot of very nice libraries like pandas which will allow me to read and write the .csv file and perform transformations on it and MLP which will allow me to create some quick validation plots BEFORE i decide to implement them in my webapp. This will just save me a lot of time. Thankfully i have all of the data stored locally so i dont need to hassle too much with API, routing or other data fetching. I can just focus on creating a good system.

References

- [1] R. Roy, "Air pollution forecasting - lstm multivariate," 2022, accessed: 2025-12-04. [Online]. Available: <https://www.kaggle.com/datasets/rupakroy/lstm-datasets-multivariate-univariate>